CS 185/285 Assignment 3: Q-Learning and Actor-Critic Algorithms

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Spring 2026

1 Experiment 1: Deep Q-Learning

1.1 DQN

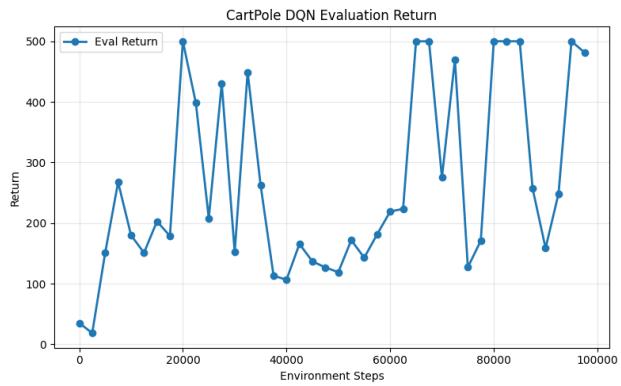

Figure 1: CartPole result with DQN

1.2 Double Q-Learning

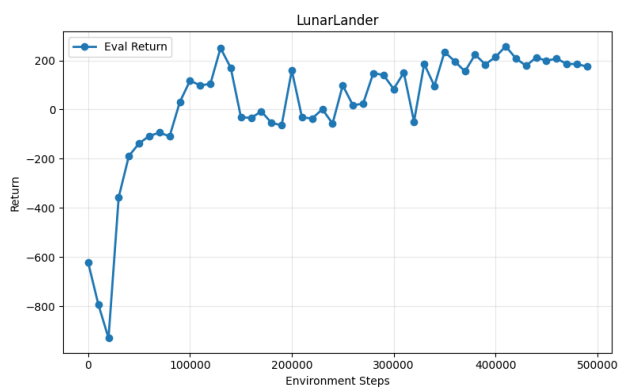

Figure 2: LunarLander result with double-Q DQN

Fig. 2 and Fig. 3 shows results of training with double-Q DQN. Both experiments are based on the default parameters.

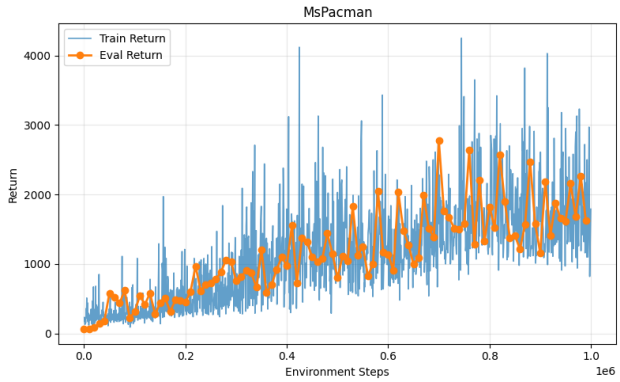

Figure 3: MsPacman result with double-Q DQN default setting.

Why is the training return and eval return different early in training? The train and eval returns differ because they are measured under different policies. Training return is collected from the epsilon-greedy exploration policy, while eval return is measured using the greedy policy with no exploration. Early in training, the greedy policy is based on poorly learned Q-values and can perform very badly, whereas random exploration in MsPacman can achieve nontrivial reward by chance. Later in training, eval matches train because the learned greedy policy stops making exploratory mistakes, but training still uses epsilon-random actions.

1.3 Experimenting with Hyperparameters

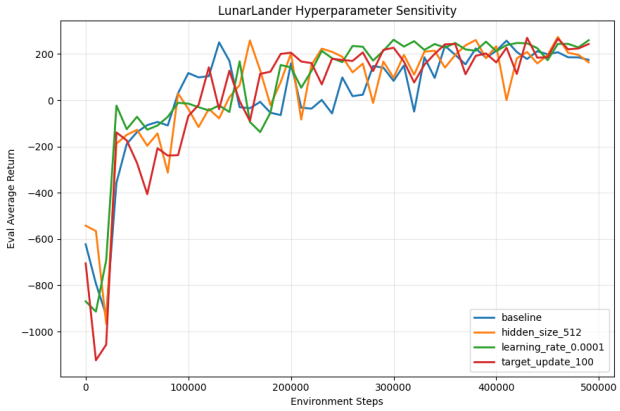

Figure 4: Results of different hyperparameters for LunarLander experiments. Each variations were selected to outline the importance of each parameter. `lr_0003` lowers the learning rate to show the optimization stability. The `target_update_250` tests how stale the target network should be. More frequent updates can make the learning faster, but they can also make the bootstrapped target less stable. The `hidden_size_512` increases the network from 2×256 to 3×512, to test the model capacity.

Fig. 4 shows results of training LunarLander with different hyperparameters. The used hyperparameters are outlined in Table 1. Despite the different hyperparameters, the runs at the end of learning gained similar eval returns.

Table 1: Hyperparameter configurations for LunarLander. All the parameters that are not noted in this table are the same as the baselines.

Exp Name	hidden_size	num_layers	learning_rate	target_update_period
`baseline`	256	2	0.001	1000
`target_update_250`	256	2	0.001	250
`lr_0003`	256	2	0.0003	1000
`hidden_size_512`	512	3	0.001	1000

2 Experiment 2: Soft Actor-Critic (SAC)

2.1 SAC

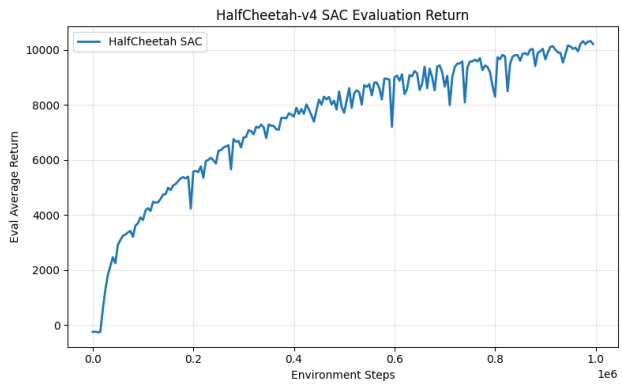

Figure 5: HalfCheetah results with SAC.

2.2 Automatic Temperature Tuning

Does auto-tuning improve or achieve comparable performance to the fixed temperature? Auto-tuning does not outperform the fixed temperature in this run, but it achieves comparable performance as shown in Fig. 6. The fixed-temperature run reaches about 10205 eval return, while the auto-tuned run reaches about 9421. Fixed alpha = 0.1 is still better here, but auto-tuning is clearly competitive.

How does the temperature evolve during training? Does it increase, decrease, or stabilize? The temperature in the auto-tuned run decreases at the beginning, then increases, and finally stabilizes. Concretely, it starts near 0.1, drops to about 0.03 around 50k steps, then gradually rises and slowly settles around 0.13 later in training.

Why might the temperature change in this particular way for HalfCheetah? For HalfCheetah, this pattern makes sense because early in training the policy is already very random, so SAC reduces the temperature since extra entropy is unnecessary. Later, as the policy becomes more deterministic while learning a good running gait, the algorithm increases the temperature to maintain the target entropy and preserve enough exploration. Once the policy's entropy is near the target level, the temperature stabilizes.

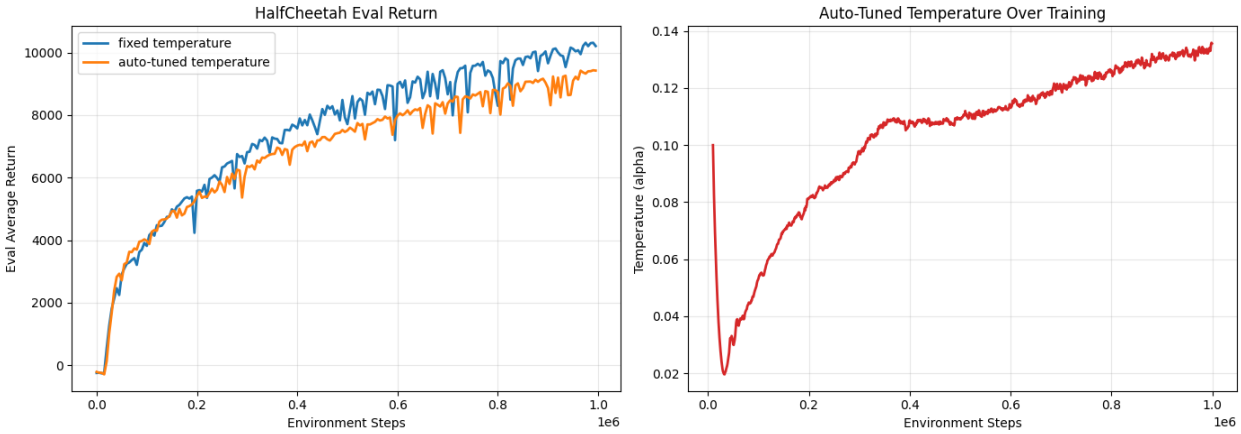

Figure 6: Comparison of HalfCheetah with fixed temperature and auto-tuned temperature.

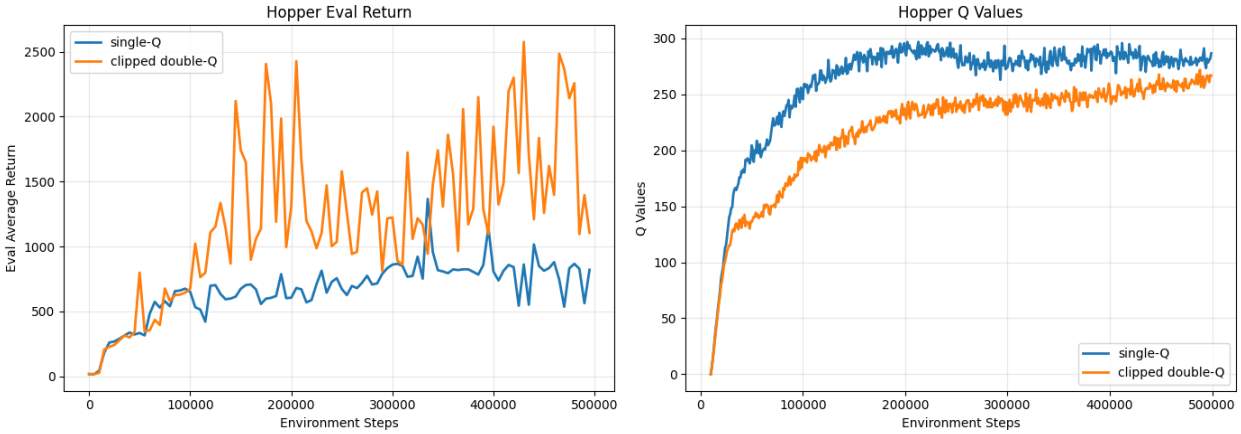

Figure 7: Result of Hopper with single-Q and clipped double-Q.

2.3 Stabilizing Target Values

Which one works best? How does the results relate to overestimation bias? Clipped double-Q performed significantly better than single-Q on Hopper, as shown in Fig. 7. The Q-value plots are consistent with overestimation bias: single-Q learned larger Q-values than clipped double-Q, but those higher value estimates did not correspond to better returns. This suggests the single-critic method was overestimating action values, whereas clipped double-Q reduced that bias by using the minimum of two critics for the target, leading to more reliable learning and higher returns.